

Formula sheet for Quiz 6

Note to students: you will be handed a copy with the quiz, so nothing here needs memorizing. It lists more than any one question needs, so deciding which applies is still your job.

Image tensors

A batch of images is (N, C, H, W) : batch, channels, height, width. PyTorch uses this **CHW** order; most image libraries hand you **HWC**, and matplotlib wants HWC to display. Pixel (r, c) is row r from the **top**, column c from the left.

Convolution output size

For one spatial axis, with input size W , kernel k , padding p , stride s and dilation d :

$$W_{\text{out}} = \left\lfloor \frac{W + 2p - k}{s} \right\rfloor + 1 \quad \text{with dilation: } W_{\text{out}} = \left\lfloor \frac{W + 2p - d(k-1) - 1}{s} \right\rfloor + 1$$

MaxPool2d(k) and **AvgPool2d(k)** use the same formula with $s = k$ and $p = 0$ by default. **AdaptiveAvgPool2d(m)** gives $m \times m$ whatever went in.

Parameter counts

$$\mathbf{Conv2d}(C_{\text{in}}, C_{\text{out}}, k) = C_{\text{out}} (C_{\text{in}} k^2 + 1) \quad \mathbf{Linear}(n_{\text{in}}, n_{\text{out}}) = n_{\text{out}} (n_{\text{in}} + 1)$$

$$\mathbf{Conv2d with groups } g = C_{\text{out}} \left(\frac{C_{\text{in}}}{g} k^2 + 1 \right) \quad \mathbf{BatchNorm2d}(C) = 2C \text{ learnable, plus } 2C \text{ buffers}$$

The $+1$ is the bias, present unless **bias=False**. A convolution's count does not involve H or W .

Receptive field

Going forward, with r the receptive field so far and j the jump (input pixels covered by one step of the current feature map), starting at $r = 1, j = 1$:

$$\mathbf{Conv } k, \text{ stride } s: \quad r \leftarrow r + (k-1)j, \quad j \leftarrow sj \quad \mathbf{Pool } k, \text{ stride } s: \quad r \leftarrow r + (k-1)j, \quad j \leftarrow sj$$

Normalization

Standardizing a channel, and the BatchNorm computation over a batch:

$$x' = \frac{x - \mu_c}{\sigma_c} \quad \hat{x} = \frac{x - \mu_{\mathcal{B}}}{\sqrt{\sigma_{\mathcal{B}}^2 + \epsilon}}, \quad y = \gamma \hat{x} + \beta$$

BatchNorm2d takes $\mu_{\mathcal{B}}$ and $\sigma_{\mathcal{B}}^2$ over the batch and both spatial axes, so one of each per channel. γ and β are learned, one of each per channel.

Batching

$$\text{iterations per epoch} = \left\lceil \frac{N}{B} \right\rceil \quad \text{standard error of the gradient} \propto \frac{1}{\sqrt{B}} \quad \mathbf{cosine } \eta_t = \frac{\eta_0}{2} \left(1 + \cos \frac{\pi t}{T} \right)$$

For C classes with scores $z_1, \dots, z_C$ and true class y : $\text{softmax } p_j = e^{z_j} / \sum_k e^{z_k}$, negative log-likelihood $\ell = -\log p_y$.

Notation

N batch size, C channels except in the softmax where it is the number of classes, H and W spatial size, k kernel size, p padding, s stride, g groups, B batch size, η the learning rate.